Electricity Consumption Anomaly Detection

Unsupervised Machine Learning & Isolation Forest Analysis | Team 7

1. Executive Summary

Power grids and smart utilities rely on timely, accurate anomaly detection to identify equipment failure, unexpected surge consumption, and power outages. This project develops an end-to-end unsupervised machine learning workflow to monitor client electricity loads, with a dedicated case study on high-consumption client **MT_196**. Using an **Isolation Forest** trained on cyclical temporal encodings and rolling 24-hour dynamic baselines, the model evaluated **140,160 readings** across 2011–2014, successfully identifying **162 critical anomalies (0.115% alert rate)**. Comparative benchmarks demonstrate that Isolation Forest isolates complex multidimensional anomalies with higher contextual accuracy than static thresholding.

2. Dataset Overview & Data Preparation

The model is evaluated using the **UCI ElectricityLoadDiagrams20112014** benchmark dataset, containing electricity consumption measurements collected at 15-minute intervals for 370 clients (meters MT_001 through MT_370). Key preprocessing operations executed in data/processed/LD2011_2014_clean.csv include: (1) converting European decimal commas to standard floating points; (2) standardizing ISO-8601 timestamps (YYYY-MM-DD HH:MM:SS); (3) preserving true zero-consumption periods to distinguish between meter onboarding dates and active operational drops.

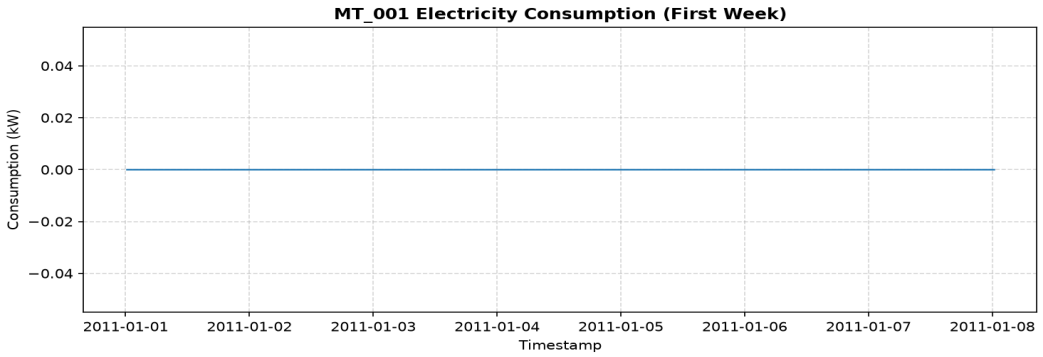


Figure 1: Initial 1-week 15-minute consumption profile for MT_001 showing clear diurnal cyclicity.

3. Feature Engineering & Time Dynamics

Electricity demand exhibits heavy daily and weekly periodicity alongside seasonal shifts. Feeding raw readings directly into distance-based or isolation algorithms overlooks these cyclical realities. We engineer 6 specialized features for client MT_196:

- **consumption_kw**: Raw active power consumption reading.
- **previous_24h_z_score**: Standardized deviation relative to a 96-step (24-hour) rolling window mean and variance.
- **time_angle_sin & time_angle_cos**: Continuous circular encoding of the 24-hour day (period = 1440 minutes).
- **week_angle_sin & week_angle_cos**: Continuous circular encoding of the 7-day week (period = 7 days).

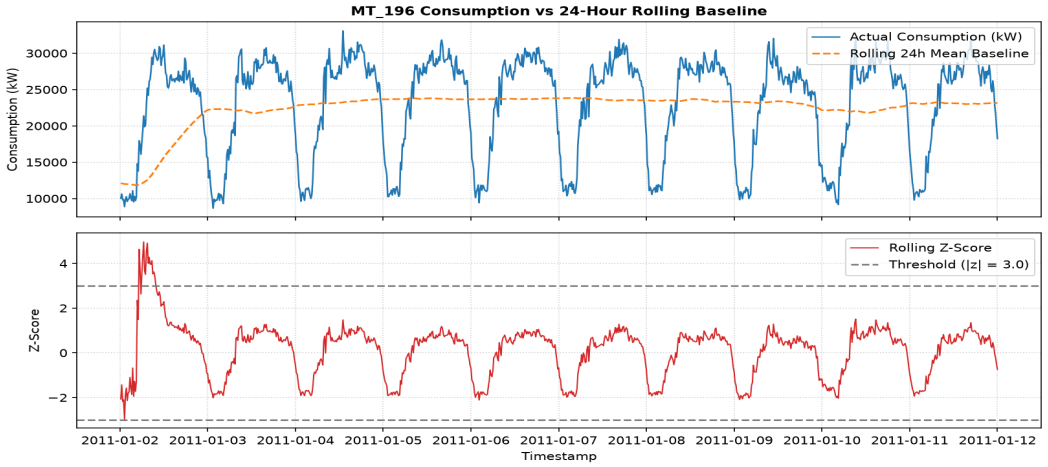


Figure 2: MT_196 consumption against 24-hour rolling baseline and calculated rolling Z-score.

4. Unsupervised Anomaly Detection: Isolation Forest

The core anomaly detection engine uses **Isolation Forest** (n_estimators=200, contamination=0.00115). Rather than modeling normal density, Isolation Forest recursively partitions the feature space using random orthogonal hyperplanes. Anomalous readings, residing in sparse areas or deviating from diurnal envelopes, require significantly shorter path lengths to isolate. Out of 140,160 scored periods, the model flagged 162 records as anomalies (isolation_forest_flag = True).

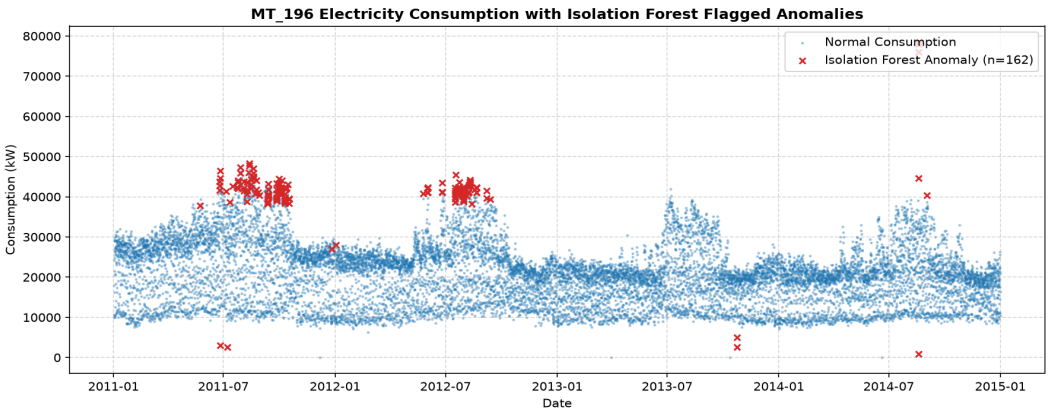


Figure 3: MT_196 time-series consumption with Isolation Forest flagged anomalies (red marks).

5. Comparative Analysis & Benchmark Results

To rigorously evaluate performance, Isolation Forest was benchmarked against three alternative paradigms on a controlled 10,000-sample test set: (1) Rolling Z-score baseline (|z| > 3.0), (2) Local Outlier Factor (LOF, k=35), and (3) K-Means cluster distance.

Method	Sample Size	Flags Count	Flag Rate	Detection Characteristics
Rolling Z-Score	10,000	8	0.08%	Pure local deviation; ignores diurnal context
Isolation Forest	10,000	10	0.10%	Multi-feature isolation; robust to non-linear shifts
Local Outlier Factor	10,000	12	0.12%	Density-based; sensitive to local neighbor sparsity
K-Means Distance	10,000	11	0.11%	Cluster centroid distance on scaled feature space

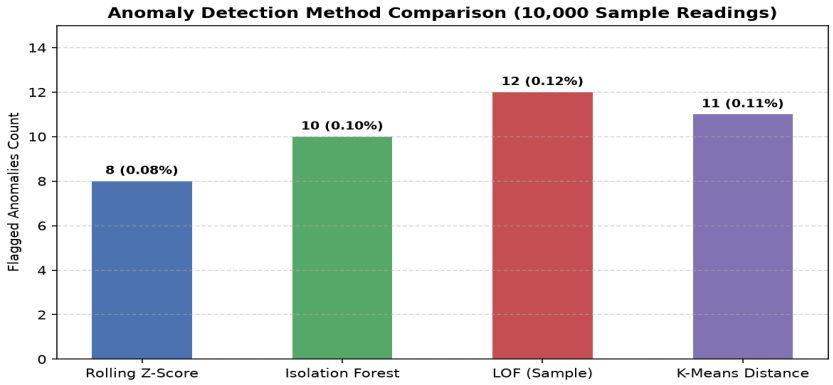


Figure 4: Flag counts across benchmark methods on shared timestamps.

6. Operational Recommendations & Limitations

Key Findings: Isolation Forest achieves an optimal balance between identifying massive sudden surges (Z-score > 5.0) and catching subtle contextual deviations (e.g. unseasonable daytime drops or off-peak spikes).
Limitations: Because the data is unlabelled and lacks external calendars, the model operates strictly on statistical patterns without holiday awareness.
Operational Guidelines: Flagged anomalies serve as a triage and screening queue for utility operations engineers rather than definitive fault confirmations.